PRACTICAL VIII

Aim: To study white-box testing and path testing, draw a Control Flow Graph (CFG) for a program, calculate cyclomatic complexity, identify independent paths, and prepare test cases.

What is White-Box Testing:

White-box testing is a software testing technique in which the internal structure, source code, statements, conditions, and control flow of a program are examined. The tester uses knowledge of the program code to design test cases.

Objectives:

- To verify the internal logic of a program.
- To identify logical errors.
- To check statements, branches, and paths.
- To improve test coverage.

What is Path Testing:

Path testing is a white-box testing technique in which test cases are designed to execute selected paths through a program's Control Flow Graph (CFG).

A path is a sequence of nodes and edges followed during program execution. Path testing helps verify different possible execution routes through the program.

Program:

The following Python program checks whether a number is positive, negative, or zero.

```
n = int(input("Enter a number: "))


if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

Control Flow Graph (CFG):

A Control Flow Graph represents the flow of execution in a program. Nodes represent statements or conditions, while directed edges represent the flow from one node to another.

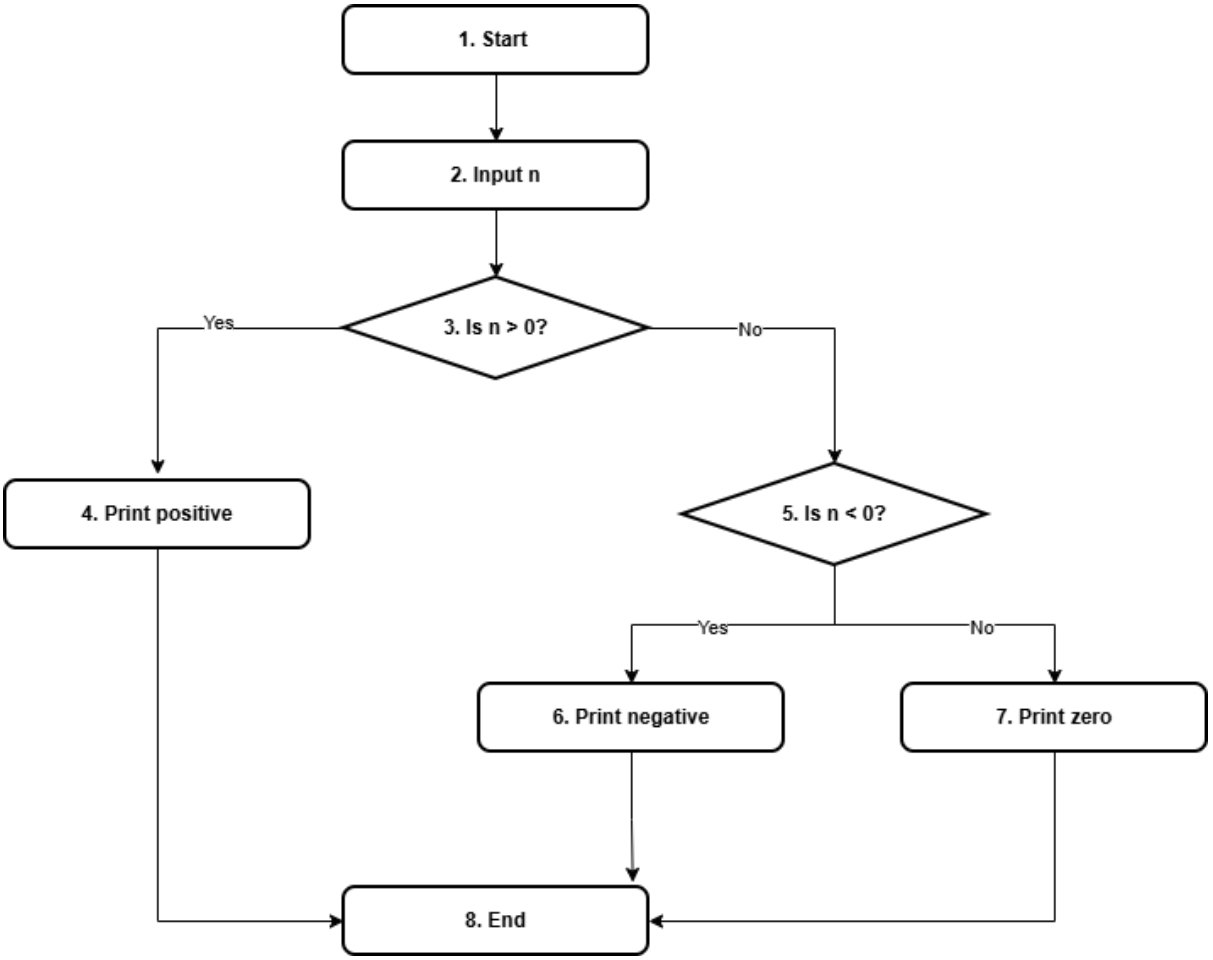
Nodes:

1. Start
2. Input number n
3. Check whether n > 0
4. Print Positive
5. Check whether n < 0
6. Print Negative
7. Print Zero
8. End

Edges:
1 → 2
2 → 3
3 (Yes) → 4
3 (No) → 5
4 → 8
5 (Yes) → 6
5 (No) → 7
6 → 8
7 → 8

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CFG:



Cyclomatic Complexity:

Cyclomatic complexity measures the number of linearly independent paths in a program.

Formula:

$V(G) = E - N + 2P$

Where:

- E = Number of edges
- N = Number of nodes
- P = Number of connected components

For the given program:

E = 9
N = 8
P = 1

Therefore,

V(G) = 9 − 8 + 2(1)

V(G) = 3

Alternatively,

V(G) = Number of decision points + 1

There are 2 decision points.

V(G) = 2 + 1 = **3**

Hence, the cyclomatic complexity of the program is **3**.

Independent Paths:

An independent path is a path that introduces at least one new edge not included in the previously selected paths.

Since cyclomatic complexity is 3, the program has three independent paths.

Path 1 — Positive number:
1 → 2 → 3 (Yes) → 4 → 8

Path 2 — Negative number:
1 → 2 → 3 (No) → 5 (Yes) → 6 → 8

Path 3 — Zero:
1 → 2 → 3 (No) → 5 (No) → 7 → 8

Test Cases:

Test Case ID	Input	Condition / Path	Expected Output	Result
TC-01	5	n > 0; Path 1	Positive	Pass if output matches
TC-02	-3	n < 0; Path 2	Negative	Pass if output matches
TC-03	0	Both conditions false; Path 3	Zero	Pass if output matches

Conclusion:

White-box testing and path testing were studied. A Control Flow Graph was constructed for the given program. The cyclomatic complexity was calculated as 3, and three independent paths and their corresponding test cases were identified.